

Google Play Store Analytics

Data Analytics Mini Project

An exploratory analysis of Google Play Store applications covering categories, ratings, installations, pricing, reviews, and update trends.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, ReportLab

Dataset: Google Play Store Apps

1. Project Overview

This project analyzes 10,357 cleaned Google Play Store app records to identify patterns in app categories, ratings, installations, pricing models, user reviews, and update activity. The analysis follows a complete data analytics workflow including data cleaning, feature engineering, exploratory data analysis, business insights, recommendations, and automated report generation.

2. Dataset Overview

Metric	Value
Analyzed records	10,357
Original columns	13
Final columns	21
App categories	33
Free apps	9,592
Paid apps	765
Latest update year	2018

3. Data Preprocessing

- Removed duplicate records from the original dataset.
- Identified and handled an invalid rating value outside the 1–5 rating range.
- Filled missing categorical values using the mode.
- Filled missing rating values using the median rating.
- Converted Reviews, Installs, and Price into numeric formats.
- Removed one malformed dataset record that caused column-shifted values.
- Converted app sizes into MB and handled 'Varies with device' values using median imputation.
- Converted Last Updated into a datetime format.
- Created additional analytical features including Year, Month, Is_Paid, Install_Category, Price_Category, Review_Install_Ratio, and Days_Since_Update.

4. Key Business Insights

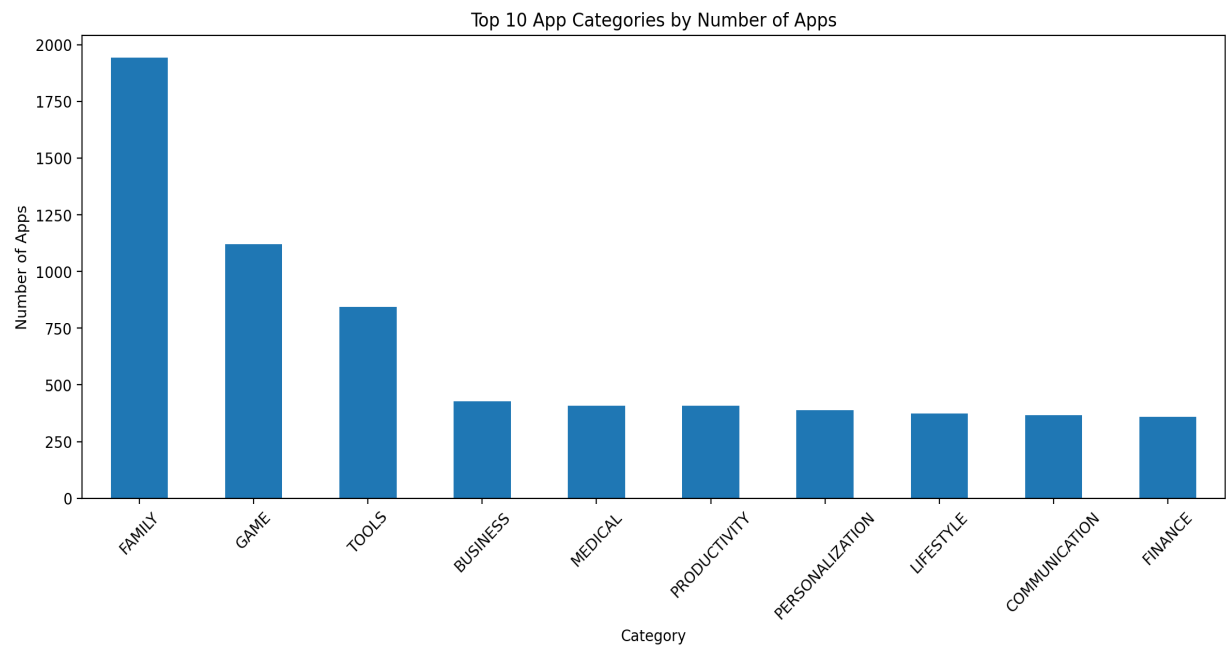
- **Largest category:** FAMILY with 1,943 app records.
- **Highest total installs:** GAME with 31,544,024,415 total installs.
- **Highest average rating:** EVENTS with an average rating of 4.40.
- **Free vs paid:** The dataset contains 9,592 free apps and 765 paid apps.
- **Average installs:** Free apps have 15,279,680 average installs, compared with 90,491 for paid apps.

- **Average rating:** Free apps have an average rating of 4.20, while paid apps have 4.27.
- **Reviews and installs:** The correlation is 0.635, indicating a moderate positive association.
- **Update distribution:** 2,018 contains the largest number of app records by Last Updated year, with 6,934 records.

Note: The year analysis represents the distribution of records by their Last Updated year and should not be interpreted directly as app creation or overall Google Play Store growth.

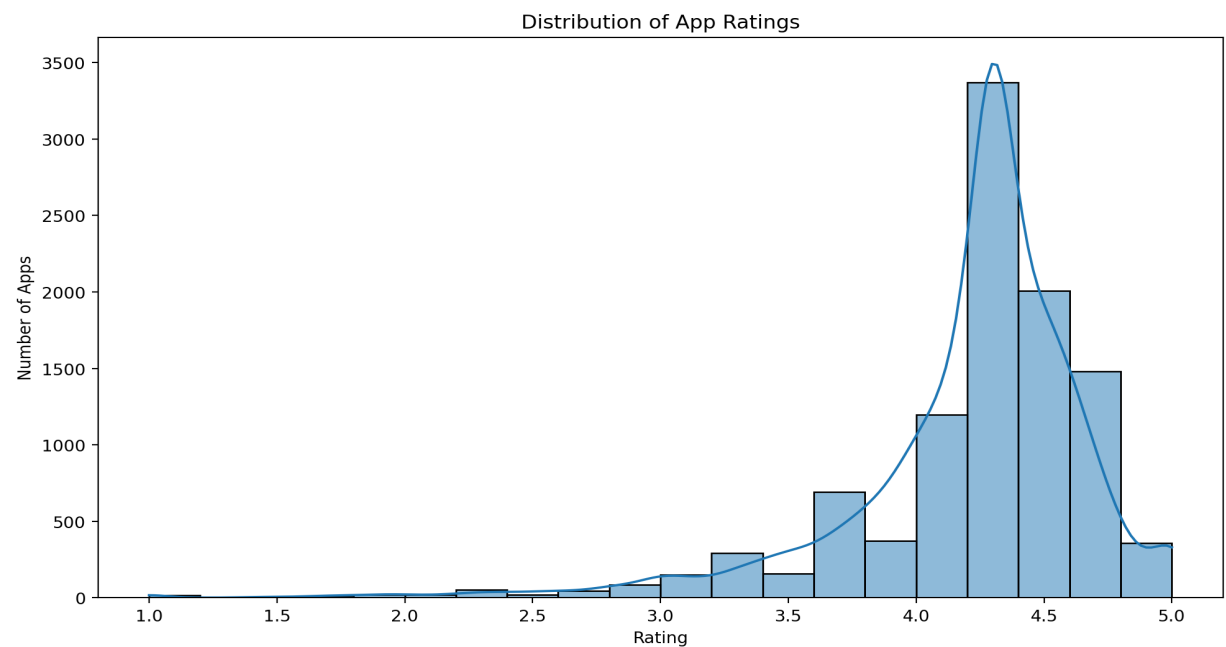
5. Exploratory Data Analysis

Category Distribution



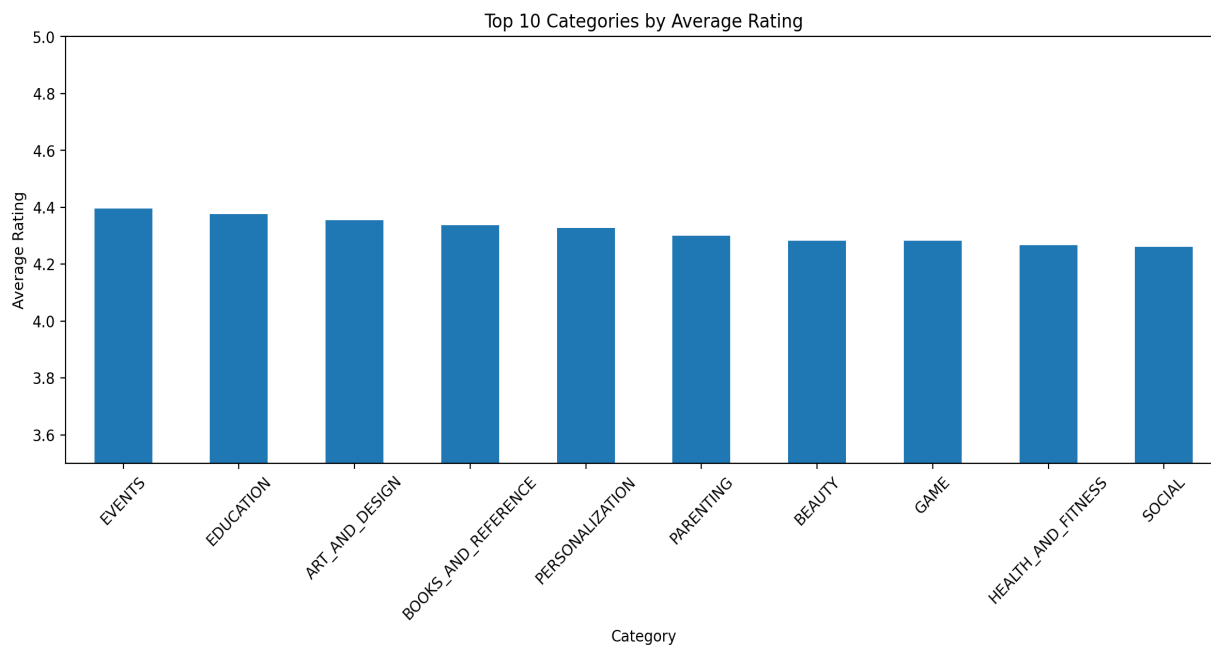
The category distribution shows which app categories contain the largest number of records.

Rating Distribution



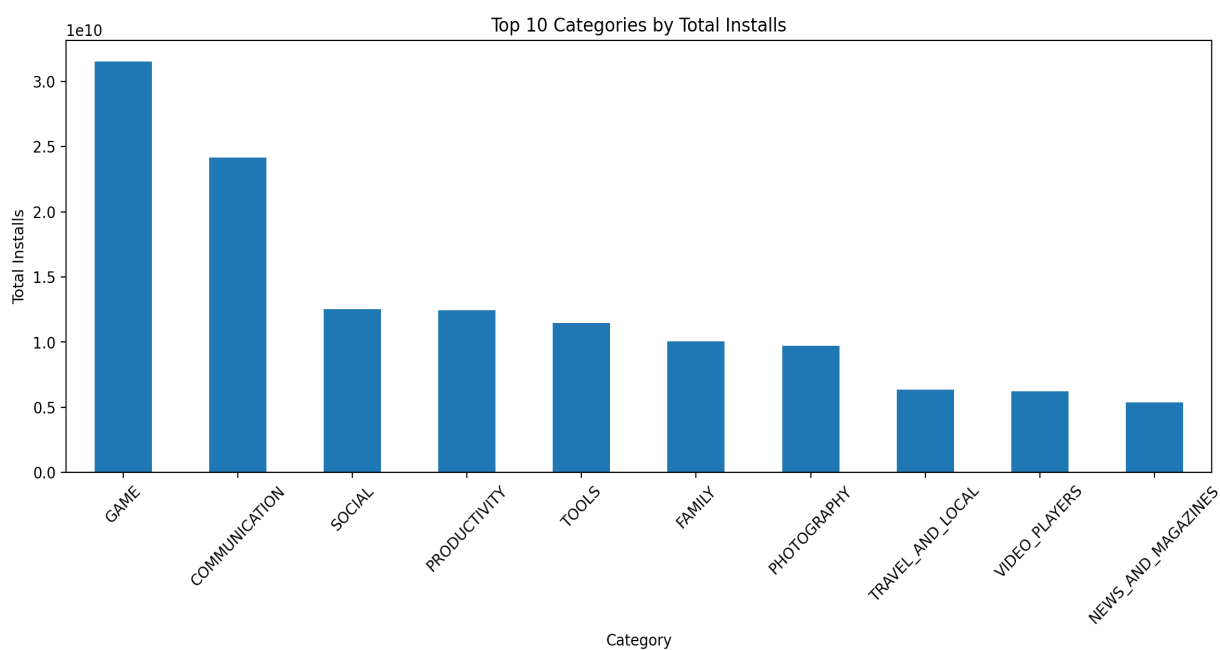
The rating distribution summarizes the overall pattern of user ratings across the dataset.

Average Rating by Category



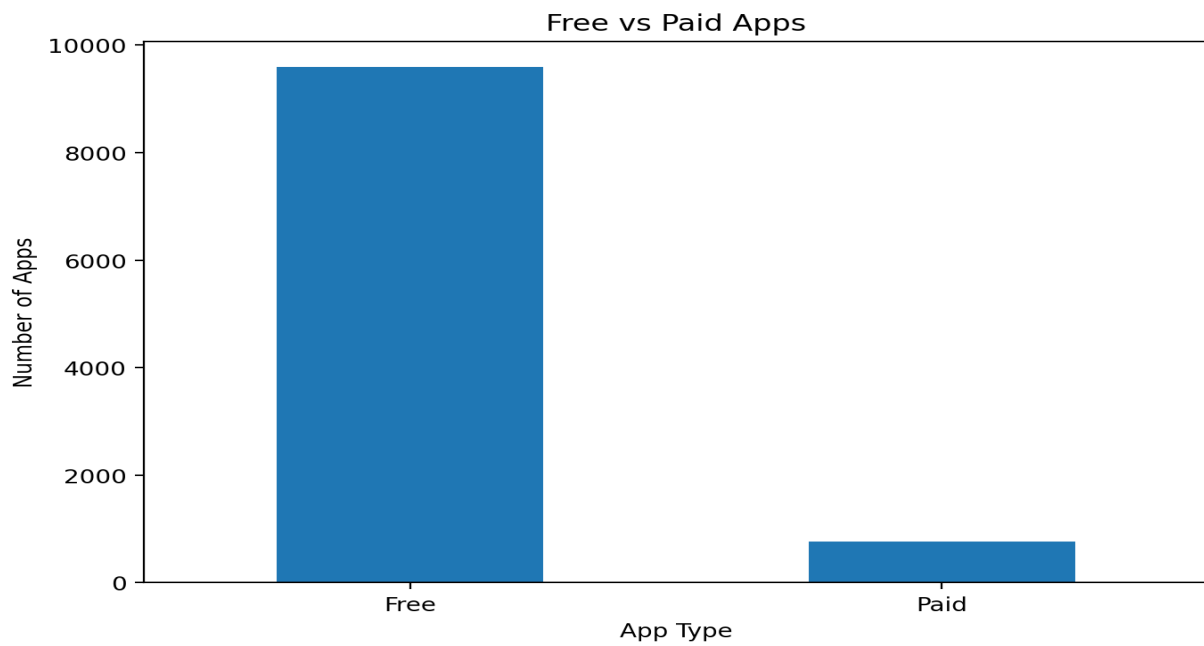
This comparison highlights categories with higher average ratings.

Installs by Category



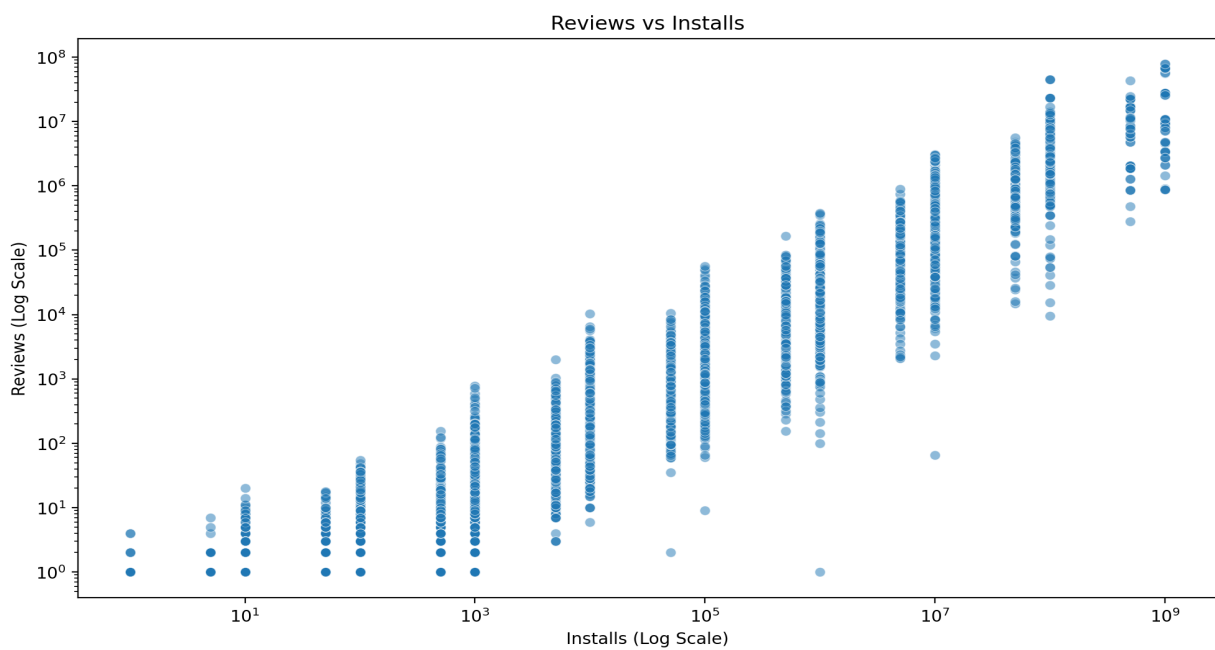
The chart compares total installs across the leading app categories.

Free vs Paid Apps



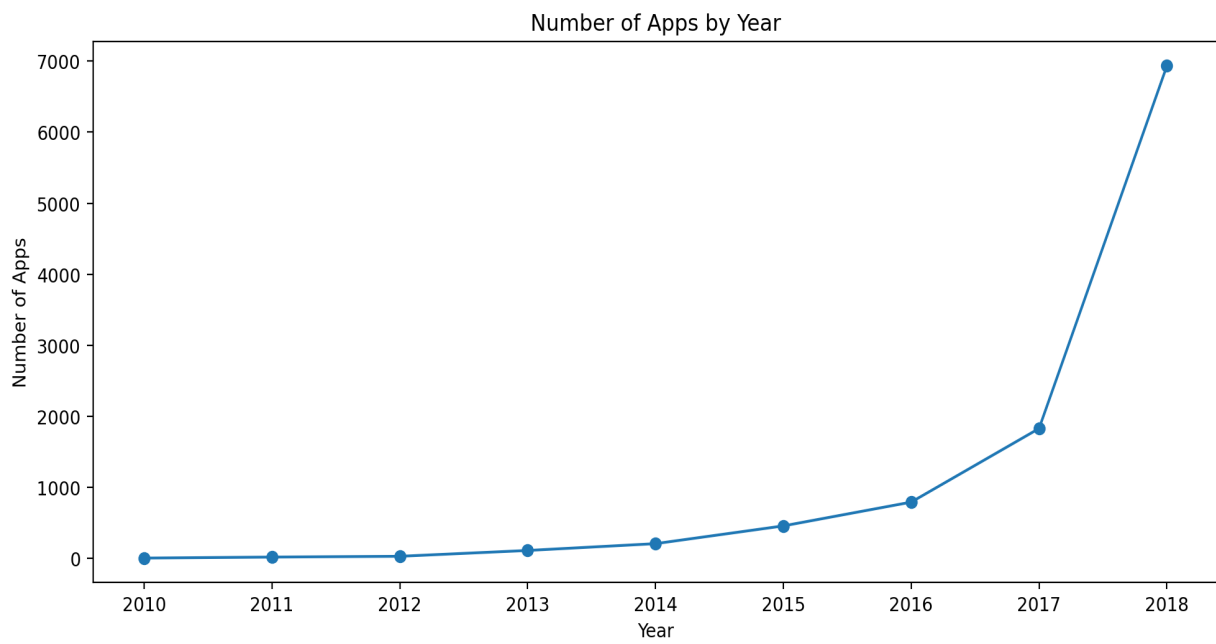
The comparison shows the strong dominance of free applications in the dataset.

Reviews vs Installs



The scatter plot examines the relationship between app installations and user reviews using logarithmic scales.

Apps by Year



This chart shows the number of dataset records according to their Last Updated year.

6. Business Recommendations

1. Focus market analysis on high-reach categories such as GAME, COMMUNICATION, and SOCIAL.
2. Prioritize user engagement and review generation because installations and reviews show a positive association.
3. Consider free or freemium strategies when targeting broad user adoption, given the dominance of free apps in the dataset.
4. For paid applications, clearly communicate differentiated features and value.
5. Monitor highly rated categories such as EVENTS, EDUCATION, and ART_AND_DESIGN for potential opportunities.
6. Maintain regular application updates to support continued relevance and user engagement.

7. Conclusion

This analysis examined 10,357 cleaned Google Play Store app records to understand category distribution, ratings, installations, pricing, and user engagement. The analysis shows that FAMILY is the largest category by number of app records, while GAME has the highest total installs. The dataset is dominated by free apps, with 9,592 free applications compared with 765 paid applications. Paid apps have a slightly higher average rating, while free apps have substantially higher average install counts. The correlation of approximately 0.635 between reviews and installs indicates a moderate positive association. Overall, the project demonstrates how data cleaning, feature engineering, exploratory analysis, visualization, and automated reporting can be combined to produce actionable business insights.